Final Year Project Logbook

Week	Date	Activities
1	21/09/2020	I searched on Google for applications of ANNs in fluid dynamics. I found many types of applications so I had to collate the tabs to I could compare them.
	22/09/2020	After consulting my supervisor on the input and output of the ANN, I searched the internet for the terms to see if a similar model had been created before. I spent the night obtaining irrelevant results as I had to understand the concept.
	23/09/2020	After researching the concept of the ANN, I finally found the correct key words to type to obtain appropriate papers, and saw some previous attempts in predicting pressure coefficients. I read through them to build familiarity with the processes involved.
	24/09/2020	I started to come up with an aim for my project. It involved the software I was planning to use, the project title, and the processes involved to achieve the desired result. This made it unnecessarily long, even though I kept it as concise as possible. After thinking of some ideas for the objectives, I wrote them down to develop them for the next day.

	25/09/2020	I managed to vaguely outline the objectives I would have to achieve to complete my project. They were good enough to keep me on track, but not enough to work from independently.
2	28/09/2020	I made my objectives more specific by adding further detail to each process, making it specific to the use of XFoil and its output data.
	29/09/2020	I had an email discussion with my supervisor to confirm that I had covered all corners for my project registration, to which he replied that I only had to show a brief understanding at this stage hence not to worry. I still painstakingly reworked the aim and objectives for another night.
	30/09/2020	I began my project registration on Moodle. My aim and objectives were not as accurate as they could have been but I decided to call it a day. When I realised my aim was
	01/10/2020	I finished registering my project on Moodle, refining the content to be more concise and specific. This happened to make the aim unnecessarily long but I couldn't be specific enough with less words. I submitted my registration.
	02/10/2020	Focusing on the other aspects of my project, I arranged a meeting with my supervisor the following week. As I have little experience in ML, a few days were spent

		understanding and researching the types of networks used. These include CNNs, RBMs, DBNs and even basic ML models such as regression.
3	05/10/2020	I found out the complexity of certain architectures meant that I could not use the latest technology for my first project as I had to understand the mechanisms of each Neural Network before combining them.
	06/10/2020	I learnt that there are different types of machine learning, unsupervised, semi supervised, supervised and reinforcement learning. At this point I watched more YouTube tutorials to understand the concepts of each type. I found the videos had low credibility as in the comments, there were people suggesting errors. I decided to come back to this topic when I have more time.
	07/10/2020	After researching the beneficiaries of the project, I found there was a lot of industries that can benefit over the use of pressure coefficient prediction, from civil to aerospace engineering. The prediction methods can save the end user millions, as a large wind tunnel program costs around £1,650,000 and a CFD program costs circa £36,000,000.
	08/10/2020	I discussed the reason why we used the input and output files that we did with my supervisor during our meeting. My supervisor explained these are the inputs used by XFoil to obtain the pressure coefficients needed. I took note of this for when I have

		more experience in XFoil and then began to familiarise myself with the software I would be using. I downloaded XFoil and began watching tutorials on YouTube to learn its functions.
	09/10/2020	I became used to using XFoil and evaluated its functions to work out which I would need. Upon looking for a tutorial, I found there was limited support due to using the wrong key words. I decided to return to it when I had more time.
4	12/10/2020	Performing a PESTLE analysis, I found that there was no legislature involved in developing this prototype CNN due to the fact the data used was open source. I found that the government and ESPRC have been heavily funding AI projects and this would be beneficial to the model's scalability, provided the resources were available. The CNN will save designers time and money as it is a verification solution. If developed with enough accuracy, a CNN may be able to negate the requirement for an expensive wind tunnel program or CFD program to be written. If it can replace wind tunnel testing, with large enough of a dataset, the CNN will change the future of aerodynamics by drastically reducing the carbon emissions caused

		by testing such as the smoke used to visualize flow fields as well reducing the electricity used to upkeep the facility. From this assessment I deemed the project to be viable as it is beneficial in all corners.
	13/10/2020	I managed to load and smooth an aerofoil on XFoil. I learnt how the co-ordinate files play a part in the shape of the aerofoil. I learnt how to smooth aerofoils, but that the detail had to be low enough to be computed efficiently. I couldn't go no further yet as I hadn't learnt the functions of its menus, I continued to explore the program.
	14/10/2020	I looked into project management methodologies. The most common in the industry were Waterfall and AGILE for small projects. AGILE allows for iterations to be made between stages, if unsatisfied with the outcome. I started to write a bullet pointed list on Microsoft Word of tasks I may have to do, in preparation for my proposal. I looked on the internet for an example AI project plan so that I could get a perspective of what was expected of me. I found as the application varied, the plan also varied, as AI mixes disciplines depending on its application, from medical to radiology. I researched traditional methods of obtaining aerofoil pressure coefficients and adapted them to a modernized approach with CNNs for use in the schedule.

	15/10/2020	I started to produce the Gannt chart for my proposal. I found Microsoft Project difficult to use, and overpriced for personal use, therefore I used TeamGantt. They provide templates as well as emailing me project progress notifications. Using the schedule as a draft, I began to produce a concise and detailed project plan. A few iterations had to be made as tasks were discovered.
	16/10/2020	I followed my tutor's advice and made the aim and objectives more specific to my application by researching them further. I also made them more concise.
5	19/10/2020	Using the research, I had previously done on the background of my project, I typed up a rationale, outlining the scope of the project and the reason why it would be beneficial to society as well as having commercial viability. The proposal was then submitted on Moodle.
	20/10/2020	I was missing a lot of information on CNNs. After researching the processes used to make CNNs, I learnt that there were new skills I had to learn. These included coding using Python 3 with Anaconda, pre-processing using Scikit-Learn, creating a CNN using Keras & TensorFlow. I was overwhelmed with the information available so I looked for textbooks. I also had a meeting with my supervisor to discuss my progress

		and to let him I may take longer than I expected with the project as I had to learn so many new skills, to which he agreed and advised me to continue.
	21/10/2020	I purchased O'Reilly's Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow from the internet following good reviews and then spent the following week reading and understand the textbook and the concepts involved.
	22/10/2020	Reading first chapter of the textbook.
	23/10/2020	Reading second chapter of the textbook.
6	26/10/2020	Reading the third chapter of the textbook.
	27/10/2020	Reading the fourth chapter of the textbook.
	28/10/2020	Evaluating what I had learnt from thTextbook.
	29/10/2020	After 6 days of reading the textbook, I had a better idea of the processes I needed to undertake. I also learnt there was a lot of foundational Machine Learning content I had to cover before attempting to create my own CNN. I made a note of the topics I had to cover and briefly scanned over them. I also created a Jupyter workspace in preparation for obtaining my data.

	30/10/2020	I decided to focus on the project and researched how I will obtain the data required for my project. I found there were methods to analyse the polars, although I couldn't find anything regarding the pressure coefficients.
7	02/11/2020	I began to test out the pre-installed aerofoils on XFoil for a Lift/Drag analysis when I saw the coefficients, I needed to take note of. I decided there must be a way to automate the process. I arranged a meeting for Thursday.
	03/11/2020	In my meeting, I asked how many aerofoils I should consider in my project, to which he replied not too many but not too few. I spent the day searching for the most credible source, which was Airfoil Tools. This website provides co-ordinate files in .dat format for use in data processing. I downloaded 20 aerofoil co-ordinates to pre-process at a later date. I learnt there are symmetrical and asymmetrical aerofoils to consider and that I would need more files for my database.
	04/11/2020	I further researched how to obtain the polars and learnt that flow speed, fluid viscosity and amplification factor has an effect of the transition point of the fluid used in the flow simulation. I then observed the effect of different angles of attack on the aerofoil and noticed the point of convergence changes at different speeds, and is corrected by the angle of attack.

	05/11/2020	After researching the effect of increasing Reynolds numbers against boundary thickness, I decided I would choose a low Reynolds number model (Re – 12500) as it would allow better identification of the optimal angle of attack at high flow speed.
	06/11/2020	I simulated the same aerofoil multiple times at increasing speeds and observed the effects upon the lift to drag ratio. I noticed at high speed, a smaller range of angles of attack were necessary to achieve convergence. I chose a speed of Mach 0.85 for my CNN.
8	09/11/2020	I learnt that the amplification factor had to be set to a level that would be preferred by the end user for use. Following the XFoil user guide, I used an Ncrit of 9 to mimic the conditions of an average wind tunnel.
	10/11/2020	I learnt how to export polars from XFoil for multiple AoAs at once using *PACC* and *aseq*. I practised obtaining polars with multiple aero foils to understand the AoAs I needed
	11/11/2020	After experimenting with various AoAs, I decided to use a range of 0° to 5° as these were most likely to converge at my chosen flow conditions.
	12/11/2020	CAE group project submission preparation.
	13/11/2020	CAE group project submission preparation.

9	16/11/2020	CAE group project evening submission.
	17/11/2020	I found that the *cpv* function produces a graphical representation of the pressure distribution, with arrow vectors. Upon reading the menu further, I found I could export the pressure coefficient for each co-ordinate in a table format for my CNN using the CPWR function. I exported some aerofoils to observe the data structure.
	18/11/2020	Upon reading further into the topic, I learnt that CNNs can understand images and I decided I will use a computer vision algorithm to allow the CNN to view the aerofoils in image format, following smoothing on XFoil. I discussed this with my supervisor during my weekly meeting. He asked me to understand the pre-processing required and undertake the work if I was prepared. I then began to research methods of visualising the data within the CNN. I learnt that images have to be a specific pixel ratio for input to output depending on the network specification. I read more research papers about CNNs that use computer vision to evaluate aerofoils to familiarise myself with the techniques used.
	19/11/2020	XFoil uses the "type I" function to select the constant parameters for calculations. Type 2 is selected as it corresponds to an aircraft in level flight at a given altitude allowing predictions to be made for less drag while cruising. I spent the day researching

		which type would be the best for my application and learnt that as I don't have a variable chord length, nor do I want to simulate aircraft take-off and landing, that type 2 was the most ideal set of fixed parameters.
	20/11/2020	I made the template for my Interim presentation, using the example on Moodle for guidance and learnt I had to cover my full understanding of the project content. I decided to spend another day justifying the reasons as to why I chose the parameters I did.
10	23/11/2020	I obtained my original aim and objectives from my project proposal and reviewed them against my feedback to improve them. Having a better understanding of the tasks to be completed following extensive research, I updated the objectives to a more conclusive list. I also shortened the aim down to 1 sentence.
	24/11/2020	To show my current progress, I explained the assumptions I made for flow parameters and presented them graphically using screenshots from XFoil to assist in justification.
	25/11/2020	I produced a slide explaining the inputs and outputs of the CNN and another explaining the function of the CNN and the next steps to be taken.
	26/11/2020	I converted my written logbook to digital format, which took me 2 days.
	27/11/2020	Converting logbook to digital format.

11	30/11/2020	I wrote the review of literature for the interim presentation, outlining what it had taught me and its pros and cons.
	01/12/2020	I collected my references and began to format them for the final draft of my presentation. I also made title pages for each section and carried out formatting to make it easier to read. As I was writing the script for my voiceover, my laptop screen failed. Without a HDMI cable, I was unable to obtain my work.
	02/12/2020	I sent my laptop to have its screen wiring repaired and waited until I could continue with my presentation.
	03/12/2020	My laptop was in repairs for another day.
	04/12/2020	I received my laptop from the repair centre. I recorded my voiceover and submitted the presentation.